

EGX360 - Intelligent Stock Market Analysis and Forecasting Platform

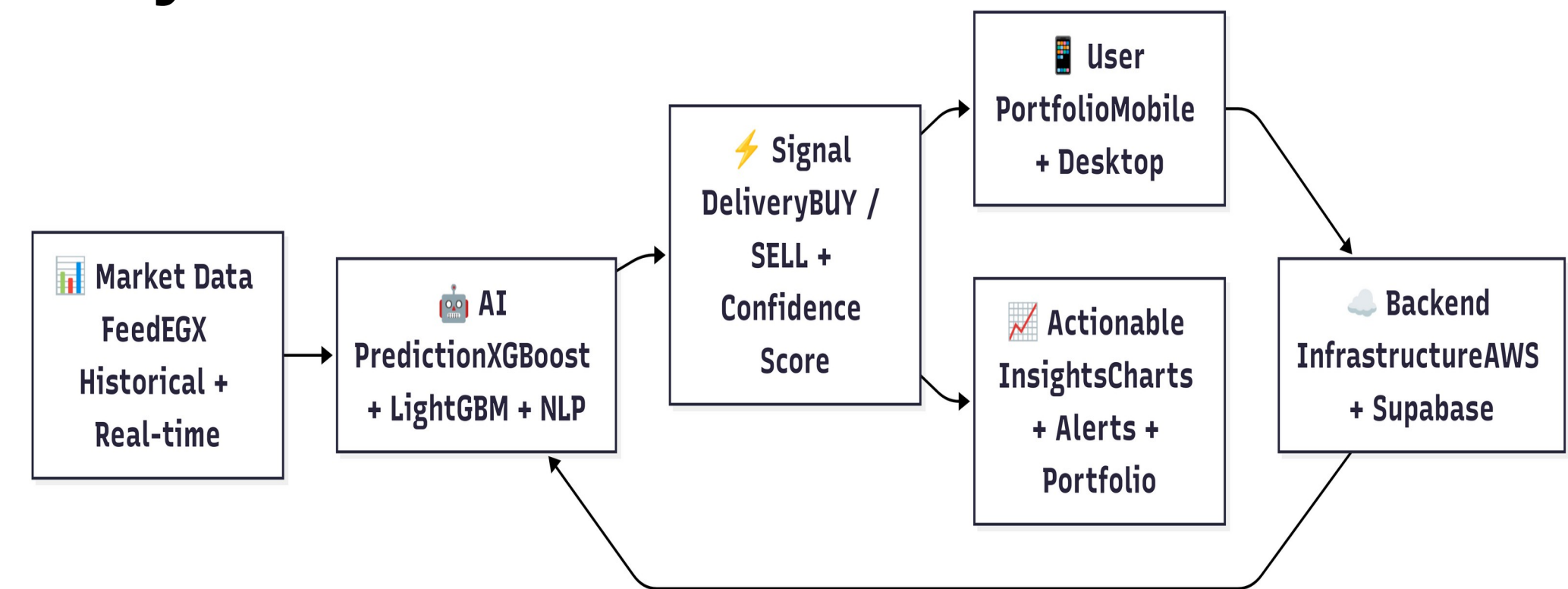
AI-Powered EGX30 Forecasting Platform



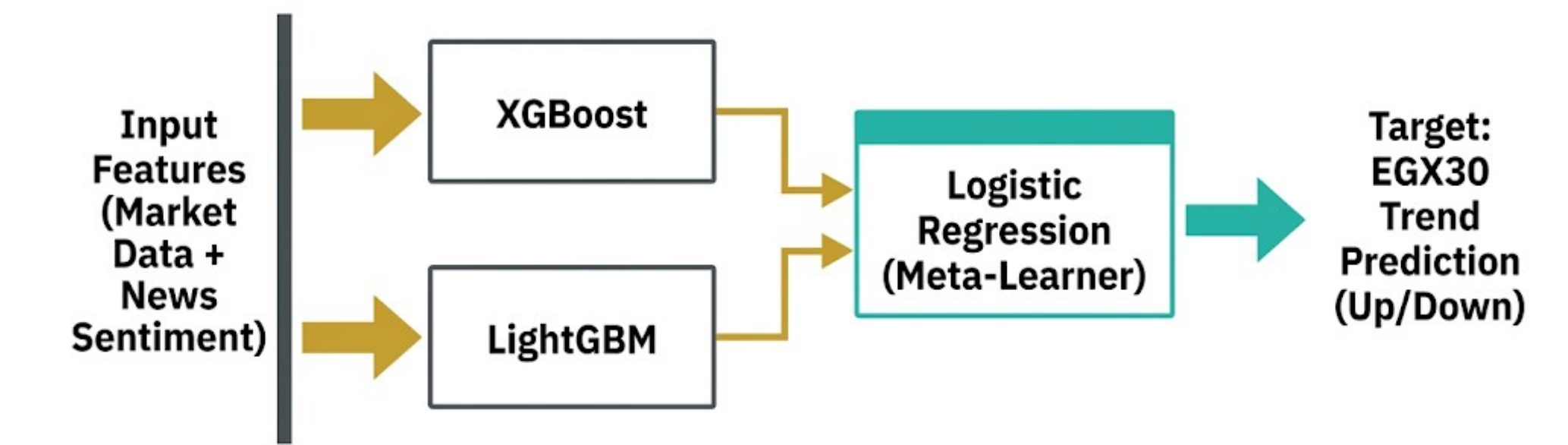
Aim& Rationale

- The Problem:** Egypt's retail investors make decisions with no real AI support.
- The Goal:** EGX360 predicts market moves before they happen — with 90% accuracy
- Rationale:** Built on AWS cloud — scales to any market, any volume, in real time.

System Architecture



Stacking Ensemble



Lessons Learned

- Technical Expertise**
 - Stacking:** Optimized XGBoost & LightGBM base learners.
 - Optuna:** Bayesian optimization for robust model stability
 - Cloud:** AWS & Supabase deployment.
- Core Knowledge**
 - Quant:** Technical & Macro correlations.
 - NLP:** Hugging Face sentiment analysis.

Conclusion

EGX360 achieves 90% accuracy by synergizing Egypt-specific macroeconomic indicators with a robust Stacking Ensemble. This project bridges the gap between raw data and actionable intelligence, providing retail investors with a production-ready tool for market volatility.a

Adaptive Dashboard



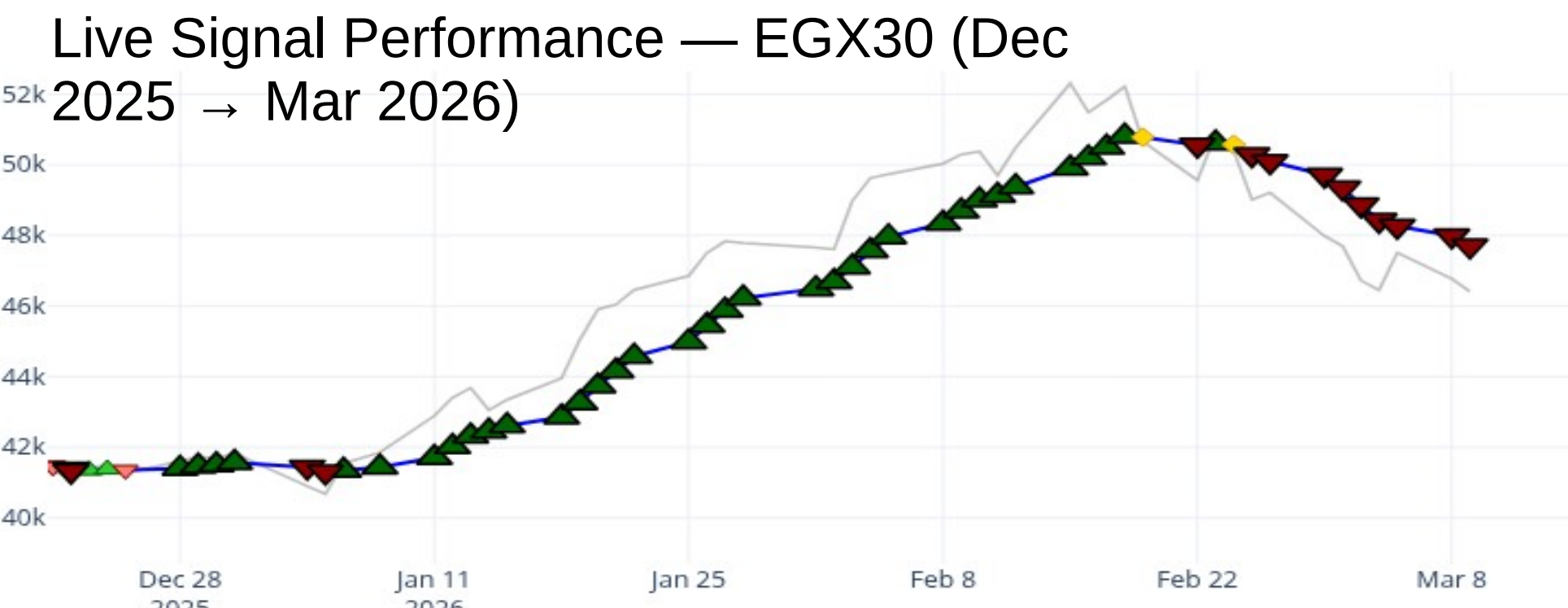
Results & Performance

+159.09%

Backtest Alpha

90%

Model Accuracy



Study	Journal / Rank	Model Architecture	Market Context	Accuracy
Livieris et al. (2025)	Springer (Q1)	SVM, LSTM, GB	55 Global Markets	51%
Park et al. (2024)	MDPI (Q2)	LSTM vs GRU	NASDAQ	65%
Chaudhari et al. (2025)	ScienceDirect (Q2)	Ensemble + Sentiment	BSE/NSE India	72%
EGX360 (This Work)	KSIU Senior Project	Stacking (XGB + LGBM)	EGX30 (Egypt)	90%

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